



EE 501: Power System Operation and Control

Course Code	EE 501
Course Name	Power System Operation and Control
L-T-P-C	3
Prerequisites	Consent of the faculty member
Intended for	UG and PG
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

An overview of power system operations and control, basic objectives of security and economics in power system operation and control, security analysis, reactive power dispatch and optimal power flows, automatic generation control, reactive power control, state estimation, brief introduction to power system restructuring and power market operations.

Text and Reference Books

Textbooks:

1. A. J. Wood and B. F. Wollenberg, **Power generation, Operation and Control**, 2nd Edition, John Wiley and Sons, 1996.
2. Prabha Kundur, **Power System Stability and Control**, Tata Mcgraw Hill Education Private Limited, 2006.
3. Loi Lei Lai, **Power System Restructuring and Deregulation: Trading, Performance and Information Technology**, John Wiley & Sons, 2001.